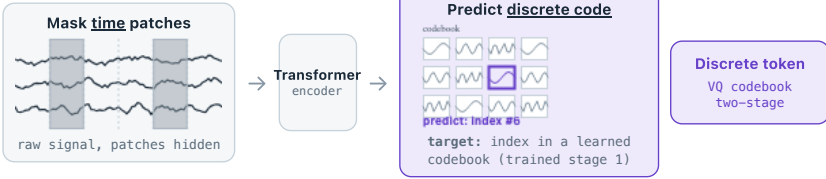


LaBraM

VQ codebook
(2-stage)

original time domain

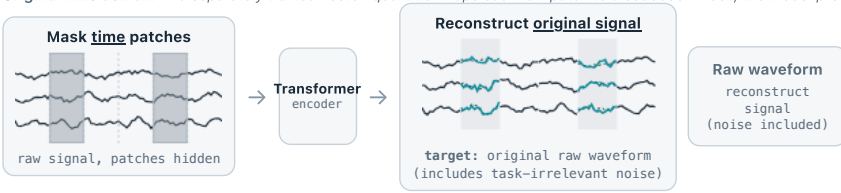


Original time domain — a separately trained vector-quantizer maps each raw patch to a codebook index; the model predicts that index.

CBraMod

CSBrain
masked
reconstruction

original time domain

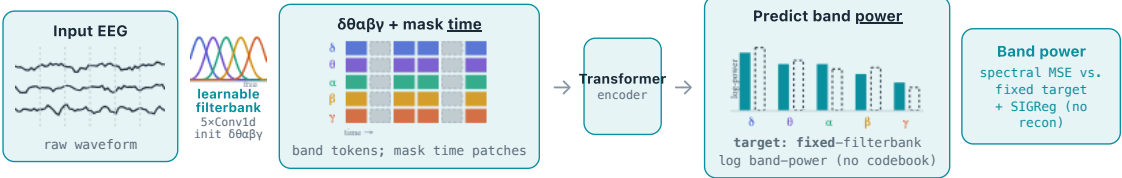


Original time domain — predict the raw microvolt waveform of masked patches; CSBrain uses the same target with cross-scale modelling.

Ours

codebook-free
spectral

frequency-band
domain



Frequency-band domain — a learnable filterbank (5 depthwise Conv1d per channel, initialised at $\delta\theta\alpha\beta\gamma$ centres) maps each patch into bands; we then predict the masked patches' log band-power against a **fixed** filterbank target. The domain (bands, not raw) and the target (power, not waveform) are what differ; the mask axis (time) is shared.

δ θ α β γ masked patch prediction target